

PROJECT 1: “Suryam Retail” – The Invisible Inventory Problem

On a busy Saturday evening, the narrow aisles of **Suryam Retail** were filled with customers navigating their weekly grocery runs. Located at the intersection of two rapidly developing residential complexes, the store had become a familiar part of the neighbourhood over the past decade. Parents stopped by after work, children picked up snacks after school, and elderly customers relied on the store for their daily essentials.

For its owner, **Mr. Rajesh Shah**, this store was more than just a business.... it was the result of years of persistence. What began 12 years ago as a modest kirana shop with a limited assortment of staples had gradually evolved into a mid-sized retail outlet. Today, Suryam Retail stocked everything from fresh produce and packaged foods to personal care and household products.

The growth had not been accidental. Mr. Shah prided himself on understanding his customers. He often stood near the billing counter, greeting regulars by name and occasionally recommending new products. Over time, this personal touch helped build a loyal customer base.

Yet, in recent months, something had started to feel different.

A Subtle Shift

At first, it was difficult to pinpoint. Footfall remained steady, and daily sales figures appeared “normal” when viewed at a glance. However, certain patterns began to emerge in small, almost unnoticeable ways.

Customers occasionally asked for products that were out of stock..... sometimes items that were known to sell quickly. On other shelves, however, products gathered dust, occupying valuable space for weeks. Staff members had begun to rearrange items frequently, trying to make better use of shelf space, but the results were inconsistent.

One evening, a regular customer casually remarked,
“I didn’t find my usual brand here last week, so I picked it up from the new store down the road.”

The comment lingered in Mr. Shah’s mind longer than expected.

Behind the Counter

Suryam Retail operated on a system that had evolved organically over the years. Procurement decisions were largely based on experience and supplier relationships. Mr. Shah maintained connections with multiple distributors, often placing orders based on what had sold well in the past..... or sometimes simply based on what suppliers recommended.

Discounts and promotional offers were introduced occasionally, especially when suppliers provided schemes. However, there was little clarity on whether these promotions actually influenced customer purchases or merely reduced margins.

The store used a basic billing system that recorded transactions efficiently. Every sale was logged, and receipts were generated without delay. Yet, beyond daily totals, the system offered limited visibility.

There was no easy way to identify:

- Which products were consistently driving revenue
- Which customers visited frequently
- Whether purchasing patterns changed over time

The data existed..... but it remained fragmented and underutilized.

Operational Realities

Inventory management was becoming increasingly complex. With hundreds of products across categories, tracking stock levels relied heavily on manual observation. Staff members checked shelves and storage areas, making judgments based on what “looked low” or “seemed sufficient.”

Occasionally, this led to over-ordering certain items while underestimating demand for others. Storage space at the back of the store was limited, and excess stock sometimes resulted in clutter, making it harder to manage incoming supplies.

At the same time, certain fast-moving items seemed to disappear faster than expected. By the time replenishment orders were placed, there was often a gap during which customers had to look elsewhere.

Mr. Shah began to wonder whether these were isolated incidents..... or part of a larger pattern.

Customer Behavior – Known and Unknown

Over the years, Suryam Retail had developed a strong base of regular customers. Many visited multiple times a week, while others came in for bulk purchases during weekends. Despite this familiarity, customer behavior remained largely intuitive rather than data-driven.

There was no formal system to distinguish between occasional and frequent buyers. Preferences were remembered informally..... often by Mr. Shah or his staff..... but not recorded systematically.

At times, new products were introduced based on trends or supplier suggestions. Some gained traction quickly, while others failed to attract attention. Without structured tracking, it was difficult to understand why.

Even pricing decisions were influenced more by market norms than internal analysis.

Mr. Shah occasionally wondered:

- Were customers sensitive to price changes?
- Did discounts actually increase sales volume?
- Were certain products purchased together more often than others?

The answers, if they existed, were hidden somewhere within the store's daily transactions.

A Growing Concern

The arrival of a new supermarket just a few blocks away had added a new dimension to the situation. While Suryam Retail still retained its loyal customers, there were subtle signs of shifting preferences.

Some customers mentioned exploring alternatives. Others reduced the frequency of their visits. These changes were not drastic enough to cause immediate alarm..... but they were noticeable.

Mr. Shah found himself relying more on instinct than ever before. Each decision..... whether to stock a product, adjust pricing, or introduce a promotion..... felt increasingly uncertain.

The Turning Point

One afternoon, while reviewing the day's sales summary, Mr. Shah remarked to his assistant:

"We have all this data... but I don't think we really know what it is telling us."

The statement captured the core of the problem.

The store was generating data every day..... through purchases, inventory movements, and customer interactions. Yet, this data was not being transformed into meaningful information.

There was a growing realization that:

- Patterns might exist, but were not visible
- Decisions were being made without structured evidence
- Opportunities for improvement might be going unnoticed

Looking Ahead

Mr. Shah has begun considering the possibility of implementing a more structured system..... one that could bring clarity without disrupting the store's daily operations.

However, he remains cautious. Any solution must be practical, easy to use, and aligned with the realities of running a busy retail outlet.

For now, the questions remain open.

Somewhere within the store's operations lies the information needed to understand:

- How products move
- How customers behave
- How decisions impact outcomes

But uncovering these insights will require more than intuition.

Your Role

You have been approached to help make sense of this situation.

The challenge is not just to organize data..... but to **interpret the business, identify patterns, and design a system that supports better decisions.**

The answers are not explicitly given.
They must be discovered.

Here are **two additional pages** that you can directly append to your *Suryam Retail* case (from your attached document). These continue the tone, deepen ambiguity, and strengthen analytical thinking without revealing solutions.

Conversations from the Shop Floor

In the weeks following his reflection, Mr. Shah began to pay closer attention to the day-to-day conversations happening within the store.

Near the billing counter, one of the staff members mentioned how certain customers had started asking for alternatives more frequently.

“Sir, this brand was not available last week also... they asked for something else this time.”

Another staff member, responsible for stocking shelves, pointed out that some products seemed to be replenished more often than expected, while others remained untouched despite being prominently displayed.

These observations were not new—but they had rarely been discussed in a structured way.

Fragments of Routine

Across Suryam Retail, information existed in fragments.

Sales were recorded at the billing counter.

Stock levels were observed visually on shelves and in storage.

Supplier interactions were based on phone calls and informal agreements.

Each activity operated smoothly within its own space. However, connections between them were not always visible.

For instance:

- A product might sell quickly—but the reason why was unclear
- A discount might be offered—but its impact was not measured
- A customer might return frequently—but this pattern was not recorded

The store functioned efficiently on a daily basis, yet the broader picture remained incomplete.

Moments of Approximation

Many decisions within the store relied on estimation.

Reordering stock was often based on how quickly items appeared to be moving.

Shelf placement was adjusted based on what “seemed” to attract attention.

Pricing changes were influenced by market trends rather than internal patterns.

These approximations had worked reasonably well in the past. They allowed flexibility and quick decision-making.

But as the number of products and customers increased, these judgments became harder to manage consistently.

Questions began to emerge:

- Were certain products selling because of their placement—or despite it?
- Were stockouts occurring due to demand—or due to timing of orders?
- Were customer preferences changing gradually, without being noticed?

The Weight of Familiarity

Much of Suryam Retail’s success had been built on familiarity.

Mr. Shah and his staff recognized regular customers, remembered preferences, and adjusted recommendations accordingly. This human connection remained one of the store's strongest advantages.

However, familiarity also had its limits.

As new customers entered the neighborhood and product variety expanded, relying solely on memory became increasingly difficult. Not every interaction could be tracked informally.

There were moments when:

- A customer's preference was forgotten
- A pattern in purchasing behavior went unnoticed
- A decision was made based on incomplete understanding

These moments were small, but they added up over time.

A Quiet Realization

Late one evening, as the store prepared to close, Mr. Shah walked through the aisles. Some shelves were nearly empty. Others were fully stocked.

Nothing appeared unusual—yet everything felt slightly out of sync.

He paused for a moment and thought:

“If we could see everything together... would it tell us a different story?”

The question did not demand an immediate answer.

But it marked a shift—from managing operations to wanting to understand them.

Between Daily Sales and Deeper Insight

Suryam Retail continues to operate much as it always has.

Customers walk in.

Products are purchased.

Suppliers deliver stock.

Transactions are recorded.

From an operational standpoint, the store is stable.

And yet, there is a growing sense that **what is visible on a daily basis may not fully explain what is happening over time.**

The Nature of the Situation

The store is not facing a sudden crisis.

Instead, it is experiencing gradual changes:

- Subtle shifts in customer preferences
- Variations in product movement
- Increasing complexity in managing inventory

These changes are not disruptive individually.
But together, they create uncertainty.

Signals Within the System

Within everyday activities, there are signals:

- Certain products consistently running out
- Others remaining unsold despite availability
- Customers adapting their choices based on availability
- Sales patterns that fluctuate without clear explanation

These signals are present—but not organized.

They exist across:

- Billing records
- Inventory observations
- Supplier interactions
- Customer conversations

The challenge is not collecting data.
It is **connecting it**.

A System Yet to Take Shape

Mr. Shah does not envision a complex or rigid system. The store's strength lies in its simplicity and personal touch.

What he seeks is something more subtle:

- A way to understand patterns without slowing down operations
- A way to make decisions with greater confidence

- A way to move beyond guesswork—without losing flexibility

What form this system should take remains unclear.

Where You Begin

You are entering Suryam Retail at a point where:

- Operations are ongoing
- Questions are emerging
- Clarity is still developing

Your task is not to impose a predefined structure.

It is to:

- Observe what exists
- Identify what is relevant
- Organize what is currently scattered

And in doing so, help the store move from:

👉 *“We are managing things as they come”*

to

👉 *“We understand what is happening, and why.”*

The Invisible Inventory Problem

Somewhere within the store’s daily operations lies a pattern—
not immediately visible,
not explicitly measured,
but present nonetheless.

Understanding this pattern may not change everything instantly.

But it may change how decisions are made.

Your Role (Extended Reflection)

You are not solving a clearly defined problem.

You are uncovering one.

The store has transactions.
The store has experience.
What it needs is **connection, structure, and insight**.

What you choose to observe—and how you choose to interpret it—will shape the outcome.

Over time, another layer of behavior begins to surface—one that is not immediately visible in individual sessions, but becomes noticeable when interactions are viewed in sequence.

A user who once explored widely begins to narrow their choices. Another who initially showed strong engagement gradually shifts toward shorter sessions. A third returns after a period of inactivity, but interacts differently than before.

Each of these shifts is subtle.

None of them, on their own, appears significant.

But together, they suggest that user behavior is not static. It evolves.

The challenge lies in understanding what drives that evolution.

There are also moments when content that once performed well begins to lose traction.

A series that attracted consistent viewership over several weeks gradually sees a decline in engagement. New users still discover it, but fewer complete it. Existing viewers do not return to it.

Nothing about the content has changed.

And yet, its position within the system seems to have shifted.

This raises a question that is not easily answered:

Is performance a property of the content itself—or of its context within the platform?

At the same time, certain titles begin to gain visibility without deliberate promotion.

They appear in user sessions more frequently. They are selected without being prominently featured. Their growth is gradual, but consistent.

These titles do not follow the expected trajectory of high-impact releases.

Instead, they build engagement quietly.

Understanding why this happens is not straightforward.

The idea of discovery becomes increasingly complex.

Some users seem to find content quickly, as if guided by a clear path. Others navigate through multiple layers before making a choice.

There are sessions where users spend more time searching than watching.

Is this exploration—or friction?

The distinction is not always clear.

There are also patterns in how users begin their sessions.

Some start with the homepage recommendations. Others go directly to search. A few return to previously watched content without browsing at all.

These entry points shape what follows.

But their influence is not always visible when interactions are viewed in isolation.

In certain cases, users appear to abandon content at similar points.

A drop-off occurs within the first few minutes of viewing. In other cases, it happens midway through an episode.

These drop-offs are measurable.

But their meaning is less certain.

Do they indicate dissatisfaction?

Mismatch of expectation?

External interruption?

Without context, interpretation remains open.

There is also the question of continuity.

A user who completes one episode does not always continue to the next. A series that begins with strong engagement may not sustain it across episodes.

Completion, therefore, is not uniform.

It varies across:

- Users
- Content
- Time

At times, the system appears to guide users effectively.

Recommendations align with preferences. Viewing sessions extend naturally. Engagement feels consistent.

At other times, the same system produces different outcomes.

Recommendations are ignored. Sessions are shorter. Users disengage sooner.

The difference between these outcomes is not immediately visible.

As discussions continue within the organization, the nature of inquiry begins to shift further.

Instead of asking what users are doing, the focus moves toward understanding sequences:

What happens before a user becomes highly engaged?

What happens before a user disengages?

These questions are not answered by single data points.

They require connecting interactions over time.

There is also an increasing awareness that:

👉 Not all patterns are stable

A behavior observed over one period may not persist.

A trend may emerge and disappear without clear reason.

This makes it difficult to distinguish between:

- Temporary variation
- Underlying structure

In one internal review, a team member presents a set of metrics showing overall growth in engagement.

The numbers are positive. Activity is increasing.

And yet, another team member asks:

“But do we know where that engagement is coming from?”

The question shifts attention from totals to distribution.

It becomes apparent that engagement is not evenly distributed across users or content.

A small set of users may contribute a large portion of activity.

A few titles may account for a significant share of viewership.

Understanding this distribution becomes as important as measuring overall growth.

There is also the possibility that some behaviors are being misinterpreted.

A user who appears inactive may still be engaged in a different way.

A user who appears active may not be satisfied.

The data reflects actions—but not always intent.

As these realizations accumulate, the nature of the challenge becomes clearer—though not simpler.

The platform does not lack information.

What it lacks is a way to interpret that information in context.

You are now looking at a system where:

- Behavior changes over time
- Content performance is not fixed
- Engagement is unevenly distributed
- Patterns are present but not stable

Your task is not to isolate a single metric or define a single problem.

It is to:

- Trace behavior across interactions
- Connect outcomes to preceding actions
- Identify structures within variability

Because somewhere within:

- What users start
- What they continue
- What they abandon
- What they return to

There is a story.

That story is not explicitly written.

It must be assembled.

And only then can the platform move closer to answering a question that remains unresolved:

What does engagement actually mean within this system?